METRIC
DOD-S-82708(OS)
21 DECEMBER 1983
SUPERSEDING
WS 14728
(See 6.3)

MILITARY SPECIFICATION

SEALING COMPOUND, PREFORMED (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of preformed sealing compound for filling voids in a guided missile rocket motor.

1.2 Classification. The sealing compound shall be of the following types, as specified (see 6.2):

Type I	Extruded tape
Type II	Extruded bead

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1218	ACS Chemicals

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material, properties, form, and dimensions.

3.1.1 Material. The material shall be a mixture of asbestos, zinc chromate and other inert extenders suspended in a nonvolatile liquid carrier which shall be capable of fitting the contours of an unacceptable void in a solid propellent rocket motor. The mixture shall be noncorrosive, nonhardening and shall have a green color.

3.1.2 Form and dimensions. Type I shall be preformed extruded tape and Type II shall be preformed extruded bead. Each type shall be supplied on easily removed paper backing. The dimensions shall be as specified in the contract (see 6.2).

3.1.3 Infrared spectrum. The infrared spectrum of the material shall conform qualitatively to Figure 1 and shall exhibit transmittance minimums only at the same wave lengths.

3.1.4 Chemical and physical properties. The chemical and physical properties shall conform to Table I.

TABLE I. Chemical and physical properties.

Property	Value	
	Minimum	Maximum
Ash content, percent	30	37
Zinc chromate content, percent	0.35	0.55
Consistency, millimeters (mm)	50	70

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3.2 Material Modification. The contractor shall notify the procuring activity of any change to the material formulation or manufacturing process, prior to shipping modified material in response to a contract involving this specification. Approval to ship such material does not relieve the contractor of repsonsibility for furnishing material that conforms to the specification.

3.3 Workmanship. The material shall be uniform in quality, consistently applied to the paper backing, and free of impurities or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of material from one manufacturer's lot produced without change in process or ingredients and offered for acceptance at one time.

4.3 Sampling. A representative sample of each lot shall be selected for testing.

4.4 Quality conformance inspection. The material shall be subjected to all the following inspections and tests for acceptance. When specified in the contract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified, all chemicals shall be ACS grade in accordance with MIL-STD-1218 and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant.

4.4.1 Infrared spectrum. Infrared spectrum shall be determined in accordance with the following:

a. Procedure: Place approximately 2 grams (g) of sample into a beaker and add 10 to 15 ml of methylene chloride. Let stand for a minimum of 10 minutes. Decant the top layer and centrifuge at high speed for approximately 15 minutes. Place some of the clear, supernatant solution on a clean sodium chloride plate and allow to evaporate until a tacky film is obtained. Adjust the film thickness to provide a spectrum where the minimum transmittance is between 2 and 20 percent at 3.4 microns. Scan the film from 2 to 15 microns on a double beam infrared spectrophotometer using a blank sodium chloride plate in the reference beam. Compare infrared spectrum with Figure 1.

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b. Report the results of a minimum of 1 dedermination as "accept" or "reject".

4.4.2 Ash content. Ash content shall be determined in accordance with the following:

a. Procedure: Ignite a suitable crucible, cool in a desiccator, and weigh to the nearest milligram (mg). Weigh 7 to 10 g, to the nearest mg, of sample material into the crucible and ignite with a Bunsen burner. Place into a muffle furnace at 800° to 850°C for a minimum of 2 hours or until ashing is complete. Remove the crucible and contents from the furnace, cool in a desiccator, and weigh to the nearest mg.

b. Calculation: Ash content (percent) = $\frac{B}{A}$ X 100

where A = original weight of sample, g
B = weight of residue, g

c. Report the average of a minimum of 2 determinations

4.4.3 Zinc chromate content. Zinc chromate content shall be determined in accordance with the following:

a. Reagents:

(1) Sodium carbonate, anhydrous

(2) Sodium nitrate

(3) Ligroin

(4) Zinc chromate, purified (Fisher Scientific Co. Z-35)

b. Solutions:

(1) Sodium carbonate solution, prepared by dissolving approximately 10 g of sodium carbonate in 500 ml of distilled water.

(2) Zinc chromate solution, prepared by adding 0.100 g of zinc chromate and 0.500 g of a mixture of 10 parts by weight (pbw) sodium carbonate and 1 pbw sodium nitrate into a suitable crucible. Fuse the mixture until complete decomposition has been effected. Leach with 30 to 40 ml of distilled water. Wet a number (No.) 42 filter paper with a small amount of hot (60° to 80°C) sodium carbonate solution. Filter the mixture through the wet filter paper into a 100 ml volumetric flask. Wash the precipitate with sufficient sodium carbonate solution to bring to 100 ml volume. Mix well.

(3) Color standard solutions, prepared by adding 0.002 g (2 ml), 0.004 g (4 ml), 0.006 g (6 ml), and 0.008 g (8 ml) aliquots of the zinc chromate solution to four Nessler tubes and diluting each to 50 ml with sodium carbonate solution. Mix well.

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c. Procedure: Place approximately 10 g of sample, weighed to the nearest mg, (weight D), into a previously weighed 180 ml centrifuge tube. Add approximately 40 ml of ligroin and allow to stand for a minimum of 16 hours. Mix thoroughly with a rod. Wash the rod with ligroin catching the washing in the tube. Add more ligroin to nearly fill the tube. Centrifuge until the pigment is settled (approximately 30 minute). Decant the clear supernatant liquid. Add more solvent, mix with a stirring rod, centrifuge, and decant for a minimum of four times. Set the tube on top of a warm oven for approximately one hour and then place in an oven at 105° to 110°C, for approximately 2 hours. Cool in a desiccator and weigh (weight E). Mix the dry pigment with a rod and place 1 to 2 g, weighed to the nearest mg (weight W), into a previously weighed crucible. Add five times the pigment weight of a mixture of 10 pbw sodium carbonate and 1 pbw sodium nitrate. Fuse the mixture until complete decomposition has been effected. Leach the melt with 30 to 35 ml of distilled water. Transfer the cake to a small beaker and agitate with a rod to entirely disperse the sample. Add a few drops of 95 percent ethyl alchol and heat to near the boiling point. Filter through a wet (see 4.4.3) (b) (2) No 42 filter paper into a Nessler tube. Wash the precipitate with sufficient hot sodium carbonate solution to bring the volume to 50 ml in the Nessler tube. Mix well. Compare the filtered sample solution with color standard chromate solution which most nearly matches the sample solution. Record the weight of zinc chromate in the matched color standard to the neareest 0.001 g (weight A).

d. Calculation: $\text{Zinc chromate (percent)} = \frac{(A)\ (E)}{(W)\ (D)} \times 100$

where: A = weight of zinc chromate in matched colored standard, g
D = original weight of sample, g
E = weight of pigment after final wash, g
W = weight of fused, washed pigment sample ,g

e. Report the average of a minimum of 2 determinations.

4.4.4 Consistency. Consistency shall be determined in accordance with the following:

a. Procedure: Weigh 25 g of sample to the nearest 0.1 g. Roll the sample into a ball. Clean two glass plates approximately 0.8 x 82 x 101 mm in size) with a suitable solvent and wipe dry. Condition the ball and glass plates for 120 ± 5 minutes at 25° ± 1° and 50 ± 5 percent relative humidity. Place the sample ball between the glass plates and apply a 4.54 kg weight evenly distributing the weight over the ball. Remove the weight after 6.5 minutes ± 10 second and before disassembly measure the greatest diameter of the sample ball which is determined by measuring the wetted area on the glass.

b. Report the average of a minimum of 3 determinations.

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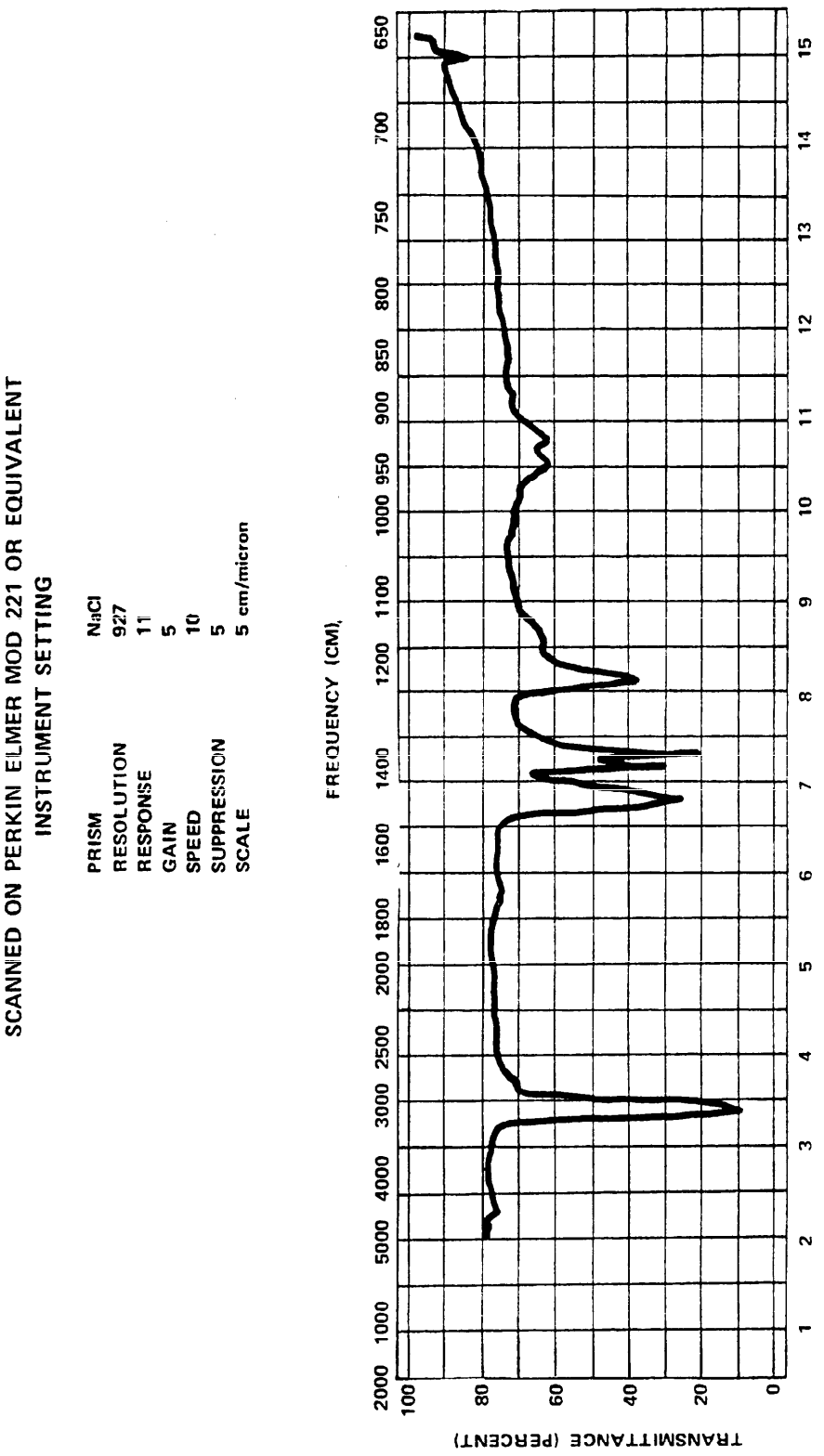


Figure 1. Infrared Spectrum of Plastic Sealing Compound Base Polymers

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4.5 Packaging and marking inspection. The inspector shall ascertain that the packaging and marking conform to the requirements of this specification by visual examination.

5. PACKAGING

5.1 Packaging.

5.1.1 Commercial. The material shall be packaged in accordance with the contractor's best accepted commercial practice.

5.2 Packing.

5.2.1 Commercial. Containers shall comply with the National Motor Freight Association Rules and the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3. Marking. Marking shall be in accordance with MIL-STD-129, and shall include, but not be limited to, the following:

a. Title, number, and revision letter of this specification

b. Manufacturer's name and location

c. Material trade name

d. Lot number

e. Type of material and quantity in container

f. Contract number

6. NOTES

6.1 Intended use. The material covered in this specification is intended for use as a sealing compound for MK 12 Mod 1 rocket motor applications.

6.2 Ordering data. Acquisition documents should specify the following:

6.2.1 Acquisition requirements.

a. Title, number and revision letter of this specification

b. Place of delivery

c. Nominal width, length, and thickness of tape desired (Type I)

d. Nominal length and diameter of bead desired (Type II)

e. Request for test results (see 4.4 and 6.2.2)

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104 (n) (2) are invoked and the DD From 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data Requirement	Applicable DID
4.4	Test Report	DI-T-2072
4.4	Certification data/Report	UDI-A-23264

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Supersession information. DOD-S-82708(OS) includes the requirements of and supersedes Naval Sea Systems Command purchase description WS 14728, dated 18 June 1973.

Preparing Activity:
Navy - OS

Project No. 8030-N081

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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